

Table of Contents

iMotions Lab > Study Setup > Study Settings and Options

Using a Test Plan for Study Design	2
--	---

Using a Test Plan for Study Design

Here you can learn about how to implement a test plan for different study designs.

Background

Test plans allow you to implement special types of study designs (i.e. a special type of randomization). Here are some examples of designs that can be implemented with a Test Plan:

Monadic Design The study can contain any number of stimuli, but each respondent will only be exposed to one stimulus.	Study consists of: STIM1, STIM2 Exposure order: RESP 1 STIM1 RESP 2 STIM2 RESP 3 STIM1 RESP 4 STIM2 ...	Completely Randomised Design Randomly orders all the stimuli for each respondent. The test plan allows for implementing any type of randomisation.	Study consists of: STIM1, STIM2, STIM3, STIM4 Exposure order: RESP 1 STIM1, STIM2, STIM3, STIM4 RESP 2 STIM4, STIM3, STIM2, STIM1 RESP 3 STIM3, STIM2, STIM1, STIM4 RESP 4 STIM2, STIM1, STIM4, STIM3 ...
Randomized Block Design Divides the group of stimuli into n homogeneous groups of size t. These homogeneous groups are called blocks. The stimuli are randomly assigned to each block.	Study consists of: BLOCK-A: A-STIM1, A-STIM2 BLOCK-B: B-STIM1, B-STIM2 Exposure order: RESP 1 A-STIM1, A-STIM2, B-STIM1, B-STIM2 RESP 2 B-STIM2, B-STIM1, A-STIM1, A-STIM2 RESP 3 A-STIM2, A-STIM1, B-STIM1, B-STIM2 RESP 4 B-STIM1, B-STIM2, A-STIM2, A-STIM1 ...	Cell Design Example: In a paired comparison test, one subsample (or cell of people) tests Product A first and then Product B, while the other subsample (cell) tests the two products in reverse order. Each cell of people can be called Cell A (A first), Cell B (B first).	Study consists of: STIM1, STIM2 Exposure order: CELL 1 - RESP 1 STIM1, STIM2 CELL 2 - RESP 1 STIM2, STIM1 CELL 1 - RESP 2 STIM1, STIM2 CELL 2 - RESP 2 STIM2, STIM1 ...

Sections

A. Create a Test Plan

1. Create the CSV File
2. Create the Stimuli Folder
3. Create a Study Based on the CSV file and Stimuli Folder
4. Deleting respondents when using testplan
5. Setting Stimuli Properties

B. Examples

- Referencing Websites
- Referencing Surveys
- Referencing both Websites + Surveys
- Referencing a Screen Recording
 - Screen Recording that Launches an Executable
 - Record anything that is on the Screen by Default
 - Launch an Application (Executable) and then Record the Screen

- Referencing a Scene Recording

A. Create a Test Plan

A Test Plan consists of:

- A CSV file containing the respondents and stimuli sequences.
- A Stimuli folder containing any files representing stimuli: images, videos, surveys, etc.

Go through the following steps to create a study in iMotions using a Test Plan.

1. Create The CSV File

Use a spreadsheet program (e.g. Microsoft Excel) to edit the Test Plan.

	A	B	C	D	E	F	G	H
1	NAME	GENDER	AGE	CITY				
2	RESPONDENT 1	F	1	NA	A_STIM1.JPG	A_STIM2.JPG	B_STIM1.JPG	B_STIM2.JPG
3	RESPONDENT 2	F	1	NA	B_STIM2.JPG	B_STIM1.JPG	A_STIM1.JPG	A_STIM2.JPG
4	RESPONDENT 3	F	1	NA	A_STIM2.JPG	A_STIM1.JPG	B_STIM2.JPG	B_STIM1.JPG

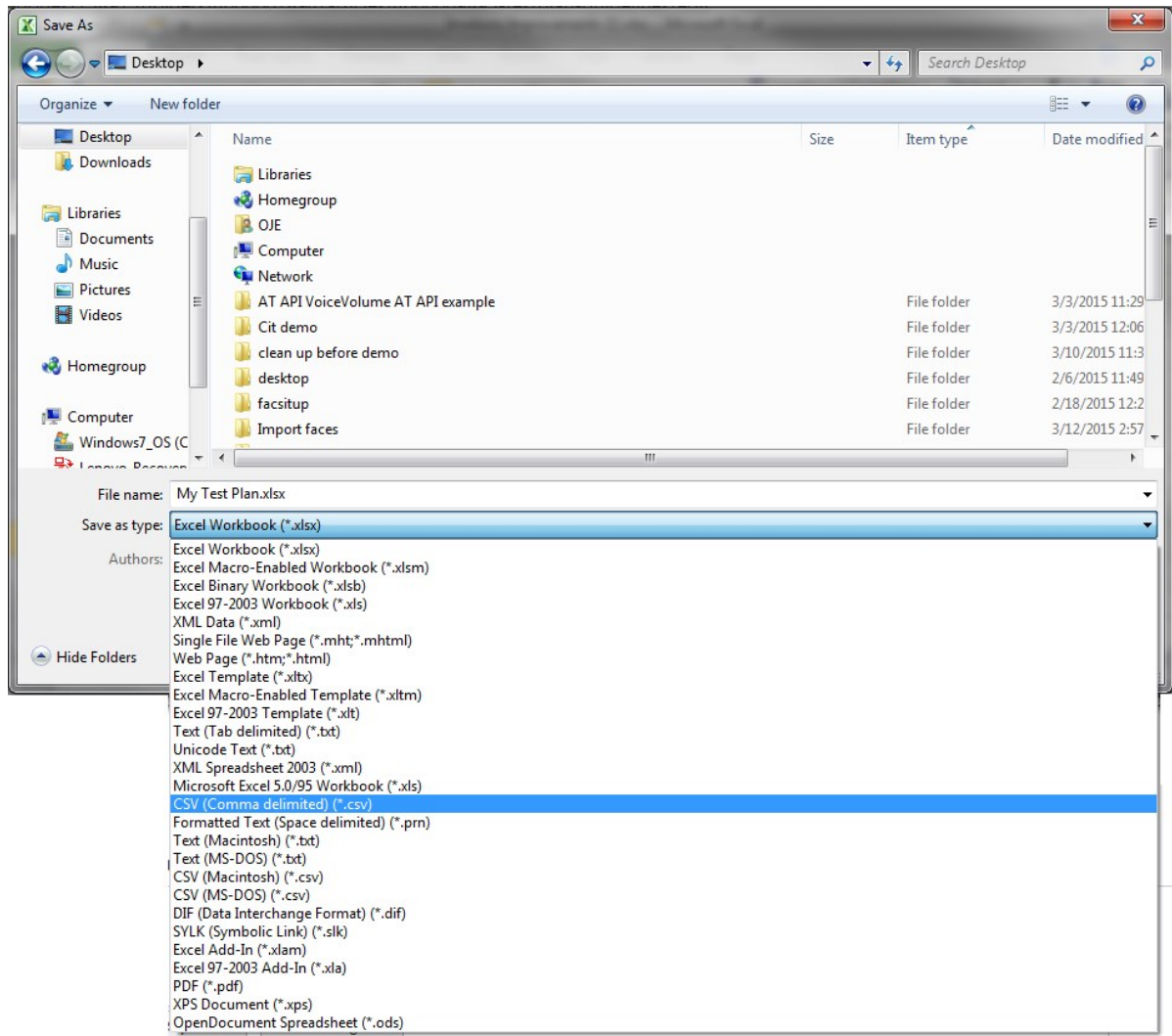
Important Formatting Considerations

- NAME, GENDER, AGE, CITY are required fields.
- The rows below these fields can be filled with default values such as shown above.
- Gender values can be either F or M.
- Age values can be between 1 and 99 (0 is not accepted as a value).
- NAME: avoid using special characters such as \ - ' @ etc. Use plain characters, e.g. Respondent 1, or RESP1,
- Avoid trailing spaces, e.g. NOT "A_STIM1.JPG ", but rather "A_STIM1.JPG"
- Avoid leading spaces, e.g. NOT " A_STIM1.JPG", but rather "A_STIM1.JPG"
- A stimulus name can only be referenced once for each respondent (i.e. once for each line in the file).

Note: It is not possible to add more respondents after you started recording your study with a testplan. Make sure you have all the necessary respondent from the beginning.

Save as a CSV File

The Excel document should be saved as a CSV formatted file.



When the document is saved the resulting file is a CSV file with the extension .csv, in this example "My Test Plan.csv".

Close Excel before continuing.

2. Create The Stimuli Folder

The Test Plan references stimuli. Often these come in the form of media files such as images, videos or survey slides.

In the example above, there are four files:

A_STIM1.JPG
A_STIM2.JPG
B_STIM1.JPG
B_STIM2.JPG

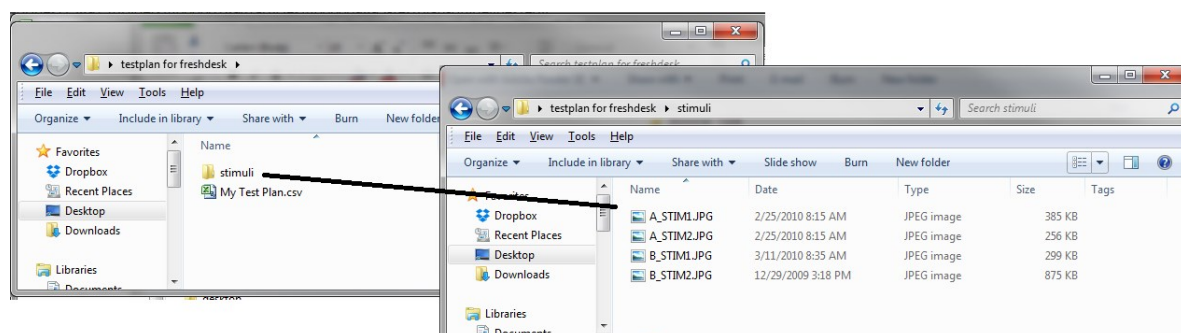
These files should all be located in a folder called "Stimuli".

Note:

- A Test Plan can not reference files that are not present in the "Stimuli" folder, i.e. files for still images, videos and survey slides - for web sites and screen recordings there are no physical files to reference.
- A folder called "Stimuli" has to be present, and the folder must NOT be empty even if you are not referencing anything (see e.g. example of a test plan referencing websites).

3. Create a Study Based on the CSV file and Stimuli Folder

At this point after completing Steps 1 and 2, you should have two items in your file system:



- A CSV file (in this example: My Test Plan.csv)
- A "Stimuli" folder, containing the stimuli referenced by the Test Plan

Note:

- the "Stimuli" folder and the CSV file should be next to each other as shown above.
- the "Stimuli" folder must be called "Stimuli" (or "stimuli").

In iMotions, create a study based on these items by using the option Test Plan, under Create a new Study

Create a new study

[Study builder help](#)

What kind of study are you planning to do?

 Screen-based Stimuli shown on a screen, (i.e. images, videos, websites, surveys, etc)	 Online Data Collection To upload and distribute to respondents online	 Glasses Using eye tracking glasses	 Screen-free recording To record a scene or face, without screen-based stimuli
 Mobile device Mobile device screen with a mobile stand	 Smart Eye Pro Smart Eye Pro system including a world model	 Virtual Reality (VR) Using a VR headset	 Test Plan Study created from a test plan

Cancel

Next

Click Browse to locate the CSV file. Select file, and click next.

Test plan study

To import a test plan study, make sure that the folder, stimuli, and csv files have all been formatted according to the test plan requirements.

For more information on how to format your test plan for import, check out this [Help Center article](#).

Select test plan file: [Browse](#)

Test plan type:

 Desktop	 Online
---	--

Note that once a test plan study has been imported, it cannot be changed in iMotions.

Back

Next

Once you have chosen the correct screen, you will be shown the stimuli and participants from your test plan. If the CSV file is correctly formatted, you will see the following verification dialog:

✕

Test plan study

Check below that the stimuli and respondents are correct

Stimuli (33)

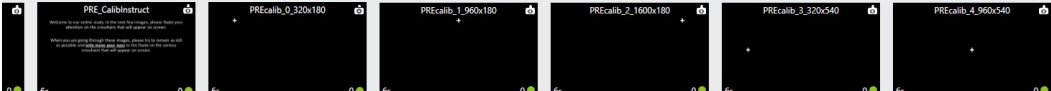


Respondents (200)

Name	City	Gender	Age	Group
S0001	CHICAGO	MALE	1	shaving
S0002	CHICAGO	MALE	1	shaving
S0003	CHICAGO	MALE	1	shaving
S0004	CHICAGO	MALE	1	shaving
S0005	CHICAGO	MALE	1	shaving
S0006	CHICAGO	MALE	1	shaving
S0007	CHICAGO	MALE	1	shaving
S0008	CHICAGO	MALE	1	shaving
S0009	CHICAGO	MALE	1	shaving
S0010	CHICAGO	MALE	1	shaving
S0011	CHICAGO	MALE	1	shaving

Back Next

- Click next to select the sensors for the study and proceed with the study. Your test plan study should have been created.



Edit stimuli
Preview study

STUDY SETTINGS SENSORS LIVE MARKERS EXPORT RECORDED DATA LATENCY SETTINGS

Study settings

Study name:

Description:

Test Plan
Study created from a test plan

Screen

Resolution: 1920 x 1080 Width: 30.9 cm Height: 17.3 cm Distance: 65 cm

Auto

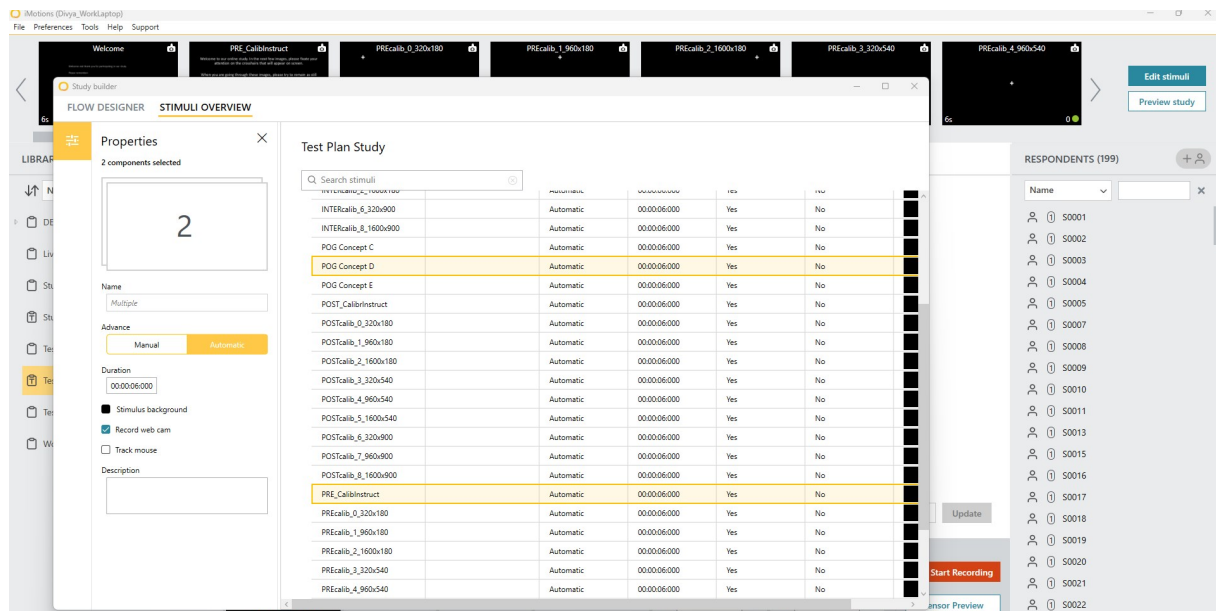
[Edit size](#)

RESPONDENTS (200) + 👤

Name

- ☒ S0001
- ☒ S0002
- ☒ S0003
- ☒ S0004
- ☒ S0005
- ☒ S0006
- ☒ S0007
- ☒ S0008
- ☒ S0009
- ☒ S0010
- ☒ S0011
- ☒ S0013
- ☒ S0015
- ☒ S0016

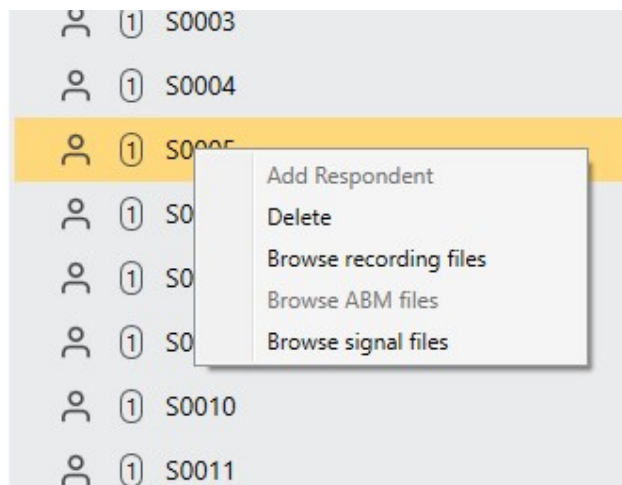
To edit the stimuli, click on edit stimuli. This will open a new window. Click on Stimuli overview to edit one or more stimuli.



4. Deleting respondents when using testplan

Starting in version 5.7 of iMotions it is possible to delete respondents when using a testplan, at any given time.

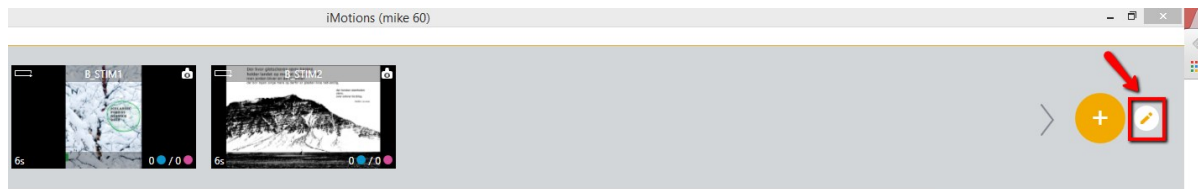
- To delete a Respondent simple right-click the respondent you wish to delete, and select delete.



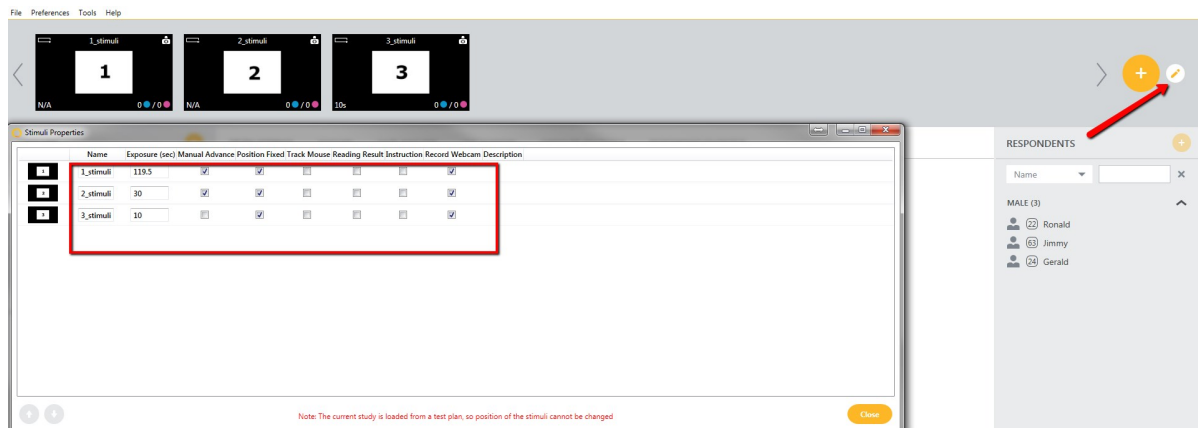
- This will remove the respondent from the study in iMotions, it will not change the content of the CSV-file. So if you import the testplan again, it will show all original respondents.

5. Setting Stimuli Properties

Use the batch editor to set stimuli properties such as exposure time, or stimuli name for all the imported stimuli.



- Clicking the button in the upper right corner opens the stimulus property editor.



Setting Exposure Time and Other Stimuli Properties In the Test Plan

See the section below, "Setting exposure time and other properties".

B. Examples

Attached to this article, you will find a zip file called TESTPLAN_document_examples.zip.

It contains a range of examples of Test Plans, including:

- A simple test plan like the one above
- A test plan for website stimuli
- A test plan for screen recording stimuli, launching an executable
- A test plan for survey stimuli

Referencing Websites

An example of a test plan that references three websites.

To define a website, the Test Plan has a header section of the form:

WebRecording;NAME; PATH-STIMULUS-ICON;URL

The arguments are described below:

NAME - The name of the stimulus that you can reference when coding the rotation for each respondent.

PATH-STIMULUS-ICON - A filepath to an icon that can be shown in iMotions to represent the web stimulus.

URL - The web address to be opened.

```
WebRecording;GooglePage;;http://google.com
```

```
WebRecording;YahooPage;;http://yahoo.com
```

```
WebRecording;BingPage;;http://bing.com
```

```
Name;Gender;Age;City
```

```
RESPONDENT1;M;26;PETERSBURG;GooglePage;YahooPage;BingPage
```

```
RESPONDENT2;M;26;PETERSBURG;YahooPage;GooglePage;BingPage
```

```
RESPONDENT3;M;26;PETERSBURG;BingPage;YahooPage;GooglePage
```

NOTE:

- The above example shows the CSV file as it would look in a normal text editor. This can also be created in a spreadsheet program, e.g. Excel.
- Even though this Test Plan does not display images or video, you will still need the Stimuli folder next to the Test Plan CSV file.
- In the header (the line that starts with WebRecording;..) there is a placeholder for an icon, in the example above the icon path is omitted - *..GooglePage;<icon path here>;http://google.com*

Above you will see two consecutive semicolons, ";;" , which means no icon is used in the Test Plan for this URL.

Referencing Surveys

A survey slide is stored on the computer as a file structure consisting of an XML file and an image folder.

If you want to include surveys in the Test Plan, follow these rules:

- Reference the survey slide as you would reference an image or video, only use the .xml extension instead.
- Put the survey slide's XML file and image folder in the "Stimuli" folder associated with the Test Plan.

Example:

```
Name;Gender;Age;City
```

```
RESPONDENT1;M;1;COPENHAGEN;instructions.jpg;buying-  
intention1.xml;buying-intention2.xml
```

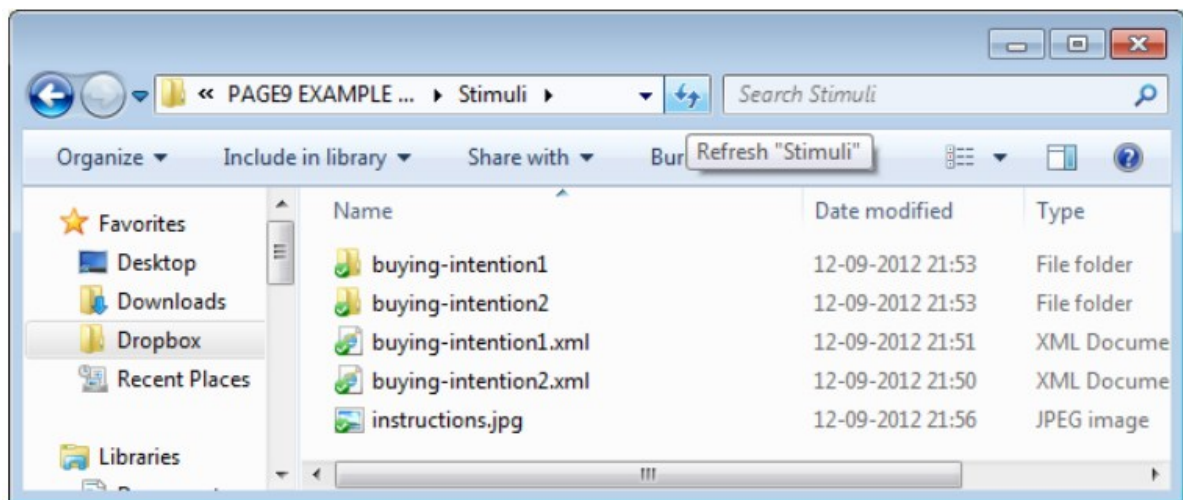
```
RESPONDENT1;M;1;COPENHAGEN;instructions.jpg;buying-  
intention2.xml;buying-intention1.xml
```

```
RESPONDENT1;M;1;COPENHAGEN;instructions.jpg;buying-  
intention1.xml;buying-intention2.xml
```

```
RESPONDENT1;M;1;COPENHAGEN;instructions.jpg;buying-  
intention2.xml;buying-intention1.xml
```

- The above example shows the CSV file as it would look in a normal text editor. This can also be created in a program like Excel.

Here is what the "Stimuli" folder contains for the example above:



Referencing Both Websites and Surveys

The example below shows a Test Plan that references websites and survey slides.

```
WebRecording;CNN;;http://cnn.com
```

```
WebRecording;BBC;;http://bbc.co.uk
```

```
Name;Gender;Age;City
```

```
CNN
```

```
RESPONDENT1;m;1;CPH;instructions.xml;CNN;surveyslide1.xml;surveyslide2.  
xml
```

```
BBC  
RESPONDENT1;m;1;CPH;instructions.xml;BBC;surveyslide1.xml;surveyslide2.  
xml
```

```
CNN  
RESPONDENT2;m;1;CPH;instructions.xml;CNN;surveyslide1.xml;surveyslide2.  
xml
```

```
BBC  
RESPONDENT2;m;1;CPH;instructions.xml;BBC;surveyslide1.xml;surveyslide2.  
xml
```

```
CNN  
RESPONDENT3;m;1;CPH;instructions.xml;CNN;surveyslide1.xml;surveyslide2.  
xml
```

```
BBC  
RESPONDENT3;m;1;CPH;instructions.xml;BBC;surveyslide1.xml;surveyslide2.  
xml
```

```
CNN  
RESPONDENT4;m;1;CPH;instructions.xml;CNN;surveyslide1.xml;surveyslide2.  
xml
```

```
BBC  
RESPONDENT4;m;1;CPH;instructions.xml;BBC;surveyslide1.xml;surveyslide2.  
xml
```

```
CNN  
RESPONDENT5;m;1;CPH;instructions.xml;CNN;surveyslide1.xml;surveyslide2.  
xml
```

```
BBC  
RESPONDENT5;m;1;CPH;instructions.xml;BBC;surveyslide1.xml;surveyslide2.  
xml
```

- The above example shows the CSV file as it would look in a normal text editor. This can also be created in e.g. Excel.

Referencing a Screen Recording

Starting in iMotions 6, you can now specify the following parameters for screen recording stimuli:

- Screen recording on/off
- Sensor recording on/off

This feature is e.g. useful when using the screen recording to launch a web based survey such as Qualtrics. Often biometric data and screen recording data is not necessary on the survey, therefore, you can now choose not to record screen and sensor data to reduce the size of the study while still being able to specify studies in testplans.

Here is an example on how to set those parameters:

```
Qualtrics,NameOfSurvey,,https://qtrial2015q4az1.az1.qualtrics.com/SE/?  
SID=SV_8wRAzRF09RETnnf
```

Screen Recording that Launches an Executable

A Test Plan can control a Screen Recording stimulus.

The Screen Recording can have two modes:

- Record anything that is on the screen by default, or
- Launch an application (executable) and then record the screen.

The Screen Recording is defined in the header using the following format:

```
ScreenRecording;<name>;<path-stimulus-icon>;<path-executable>;<INPUT-  
EXECUTABLE>
```

The arguments are described below:

ScreenRecording - The token that indicates a Screen Recording stimulus.

NAME - A unique name to use in the test plan to reference the screen recording.

PATH-STIMULUS-ICON - A file path to an icon that can be shown in iMotions to represent the screen recording stimulus.

PATH-EXECUTABLE - A file path to an executable, .exe file for a software application that will be launched at the beginning of the screen recording stimulus exposure. If this is left blank, a screen recording will still launch, recording what ever happens to be on the screen at that moment.

INPUT-EXECUTABLE - Any input argument for the executable, for example a filepath or a URL.

Record Anything that Is on the Screen by Default

A simple example with a Screen Recording.

Defines one stimulus called "Desktop"

```
ScreenRecording;Desktop;;
```

```
Name;Gender;Age;City
```

```
RESPONDENT1;M;26;MOSCOW;instructions1-1.jpg;desktop
```

```
RESPONDENT1;F;27;MOSCOW;instructions1-2.jpg;desktop
```

Launch an Application (Executable) and then Record the Screen.

Two stimuli are defined, both referencing Notepad.exe and a file that Notepad will open automatically called textfile1.txt and textfile2.txt.

This example also points to a .png file that will be used as an icon representing Notepad in the iMotions stimulus navigator.

```
ScreenRecording;NotePad1;C:\Program Files\iMotions\Attention Tool  
4\images\setup_border.PNG;C:\Windows\System32\NotePad.exe;C:\testfile1.  
txt
```

```
Name;Gender;Age;City
```

```
RESPONDENT1;M;26;MOSCOW;NotePad1
```

```
RESPONDENT2;F;27;MOSCOW;NotePad2
```

Referencing a Scene Recording

Starting in iMotions 6, the test plan feature now supports sequencing Scene Recording stimuli. If your experimental design involves a number of trials that have to appear in a certain sequence (e.g., depending on conditions or segmentation parameters), then the testplan now allows you to implement such designs. For example, a package research study where respondents must interact with a number of packages in blocks.

In order to define a scene recording in the testplan, the following line must be used:

```
SceneRecording;<Display name>[Icon file path]
```

Example 1 (with icon file):

```
SceneRecording;MySceneRecording;C:\Icons\customsceneRecIcon.jpg
```

```
Name;Gender;Age;City
```

```
RESPONDENT1;M;22;BOSTON;MySceneRecording
```

Example 2 (without icon file):

```
SceneRecording;MySceneRecording;
```

Setting Exposure Time and Other Properties

It is possible to specify optional setup parameters for the stimuli in study test plans. The parameters are defined as name value pairs, separated by white spaces.

Parameters ExposureTime and ManualAdvance are available.

A syntax example of a study test plan with parameters is given below:

```
NAME;GENDER;AGE;CITY;;;
```

```
Michael;M;22;COP;A_STIM1.JPG{ExposureTime=119.5  
ManualAdvance=true};A_STIM2.JPG{ManualAdvance=true};B_STIM1.JPG{ExposureTime=10};Desktop{ExposureTime=5}
```

```
Eric;M;43;COP;B_STIM1.JPG;B_STIM1.JPG;A_STIM2.JPG;A_STIM1.JPG
```

```
John;M;63;COP;A_STIM1.JPG;A_STIM2.JPG;B_STIM1.JPG;B_STIM2.JPG
```

```
Brad;M;24;COP;B_STIM2.JPG;B_STIM1.JPG;A_STIM1.JPG;A_STIM2.JPG
```

If the ExposureTime parameter is defined, it will override the default exposure time for the stimuli.

Defining ManualAdvance with a true parameter will set the Manual advance checkbox to true in the imported study for this stimuli. The max exposure time will be the default (30sec) unless the ExposureTime parameter is defined, then this will be used as the max exposure time.

A parameter for a stimulus needs to be defined the first time a stimulus is encountered, since the stimuli settings will overwrite all subsequent definitions of the same type of stimuli.

See the attachment TestPlanExtended.zip for an example that can be imported to iMotions.

